

ANALYSTLAB AFRICA

Experience Lab Internship Programme



HealthConnect Clinic Week 4 Project Summary

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Dataset: HealthConnect Appointment Dataset (5,000 records)

Prepared for the AnalystLab Africa Experience Lab
Week 4 Submission – Data Science Track

Table of Contents

1. Introduction
 2. The Problem This Track Will Address
 3. Resources Used
 4. Key Observations from Week 4
 5. Proposed Approach
 6. Key Considerations
 7. Proposed Focus for Week 5
- References

1. Introduction

This Week 4 Project Summary accompanies the Data Science track's Machine Learning Problem Definition Document for the HealthConnect Experience Lab. Consistent with Part 4 of the Week 4 brief, it is intended as a concise, standalone overview of this week's work and is deliberately kept brief: it summarises what was done and why, and points the reader to the main document for the underlying evidence, tables and detailed justification, rather than reproducing that material here.

HealthConnect Clinic is a fictional outpatient provider facing a recurring problem of missed appointments. From Week 4, the shared project asks how data and AI can help reduce these no-shows and improve the patient support experience. This summary sets out how the Data Science track has approached that question this week.

2. The Problem This Track Will Address

The Data Science track is responsible for defining the machine learning problem underpinning HealthConnect's no-show challenge and for assessing whether the available appointment data can support a prediction solution. Specifically, this involves proposing a target variable, identifying candidate input features, deciding how to treat cancelled appointments, and setting out an initial, evidence-informed modelling approach for Week 5.

3. Resources Used

This track's work drew on the HealthConnect Appointment Dataset (5,000 anonymised records) and the accompanying Data Dictionary. The Clinic Knowledge Base, though primarily a Generative AI track resource, was also briefly consulted to confirm what a 'Cancelled' appointment represents operationally, which directly informed the target-variable decision described in the main document. No original project resource was modified; all analysis was carried out on a working copy of the dataset.

4. Key Observations from Week 4

Three observations stood out from this week's review and are examined in full in the main document. First, the dataset is largely clean and internally consistent, with only minor and mostly structural missingness. Second, a patient's own history of missed appointments and the length of the booking lead time show the clearest early relationship with no-show behaviour, consistent with the wider clinical literature on this problem. Third, one field, `waiting_time_minutes`, shows a data-quality pattern that is difficult to reconcile with its stated definition and needs to be clarified before it can safely be used as a model input.

5. Proposed Approach

The problem has been framed as binary classification: predicting whether a scheduled appointment will result in a no-show, using only information that would be available at the time of booking. Cancelled appointments are proposed to be excluded from the primary training set, since they reflect a proactive patient action rather than the silent non-attendance the clinic is trying to reduce. A logistic regression baseline, followed by tree-based ensemble methods, is proposed for Week 5, evaluated using precision, recall, F1-score and ROC-AUC rather than accuracy alone.

6. Key Considerations

- **Data leakage:** `waiting_time_minutes` must not be used as a feature unless it can be confirmed as information genuinely available before the appointment takes place.
- **Fairness:** model performance should be reviewed across demographic subgroups before any real-world use is considered.

- **Cross-track dependency:** the target variable and feature set proposed here will need to be shared with the Machine Learning Engineering and Project Management tracks, since both depend on this problem definition.

7. Proposed Focus for Week 5

Week 5 will prioritise resolving the `waiting_time_minutes` question with the data owner, completing feature encoding and a formal statistical review of the candidate features identified this week, and training and evaluating the proposed baseline and candidate models against the metrics set out above.

References

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